Global Mg/Si and Al/Si maps of the Moon from Chandrayaan-2 CLASS with time-resolved solar excitation from GOES XRS

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Abstract

We describe a processing chain for Chandrayaan-2 CLASS X-ray fluorescence spectra that can be reproduced from public data. Each 8 s spectrum is fitted on raw counts with a Poisson likelihood and a simplified fundamental-parameter model: Sherman primary fluorescence with matrix attenuation and incidence geometry, a scattered solar spectrum, a binned Gaussian detector response, the published filter transmission and a particle-background template. The incident solar spectrum is approximated by an isothermal CHIANTI spectrum at the GOES XRS temperature of each exposure. Because solar lines carry 55–80% of the flux that excites Mg, Al and Si, element/Si ratios depend appreciably on this spectrum. We also describe an instrumental Al Kα component that appears when the spacecraft is sunlit from behind, and show that its treatment in the background model affects Al/Si. In synthetic tests Mg/Si and Al/Si are recovered with small biases, whereas Ca/Si, Ti/Si and Fe/Si are not constrained with this approach. From 1,515,239 exposures in 24 monthly archives, 578,530 spectra pass pre-declared quality criteria and give global 2° maps with quality layers and no infill. Al/Si correlates with Lunar Prospector GRS on 5° pixels ($r = 0.72$) and with nine landing-site soils ($r = 0.81$). Mg/Si correlates with GRS ($r = 0.61$), but its landing-site comparison is inconclusive. Refitting 6,327 spectra from three months with solar spectra derived from Chandrayaan-2 XSM changes the median Al/Si by −4 to +4% and Mg/Si by −21 to +14%, within the stated systematic uncertainties.

Keywords: Moon, X-ray fluorescence, Chandrayaan-2 CLASS, solar X-ray spectrum, GOES XRS, lunar surface composition

1. Introduction

The major-element chemistry of the lunar surface records how the crust formed and evolved. Highland plagioclase is rich in Al and Ca; mare basalts are rich in Fe, Mg and Ti. Global maps of these elements constrain magma-ocean crystallisation, mare volcanism and impact mixing (Heiken et al., 1991). Orbital gamma-ray spectroscopy has provided global maps of Fe, Ti, Th and K, and model-dependent estimates of Mg, Al, Si and Ca, at a resolution of hundreds of kilometres (Lawrence et al., 1998; Prettyman et al., 2006).

X-ray fluorescence (XRF) excited by solar X-rays offers better spatial resolution and direct sensitivity to Mg, Al and Si (Jenkins, 1999; Beckhoff et al., 2006). The Chandrayaan-1 X-ray Spectrometer (C1XS) demonstrated the technique during a period of very low solar activity, when only a few flares gave usable spectra (Grande et al., 2009; Narendranath et al., 2011). Its successor on Chandrayaan-2, the Large Area Soft X-ray Spectrometer (CLASS), has a geometric area of 64 cm^2 and a footprint of a few tens of kilometres (Narendranath et al., 2014; Radhakrishna et al., 2020; Pillai et al., 2021). It has operated through the rise of solar cycle 25, and its Level-1 spectra are public.

Turning XRF line intensities into composition requires the spectrum of the incident solar X-rays. The fluorescence yield of each element depends on the solar flux above its absorption edge, so a change in the solar spectrum can change line ratios such as Mg/Si or Ca/Si even when the surface composition is unchanged. The Chandrayaan-2 Solar X-ray Monitor (XSM) measures this spectrum (Mithun et al., 2020), and the instrument team's analyses combine CLASS spectra with simultaneous XSM data. For users who work only with the CLASS archive, a common simplification is to assume a reference solar spectrum or to rely on line ratios, although ratios still depend on the solar spectrum because each element samples a different part of it.

In this paper we develop and test a CLASS processing chain that can be reproduced from public data. We approximate the solar excitation of each exposure by an isothermal CHIANTI spectrum at the temperature derived from the GOES X-Ray Sensor, and we check this approximation against XSM spectra for three months. The work comprises:

1. a simplified fundamental-parameter forward model (Sherman primary fluorescence with matrix attenuation and incidence geometry, scattered solar spectrum, binned Gaussian detector response), fitted to raw counts by Poisson maximum likelihood with per-spectrum uncertainties, goodness of fit and quality flags;
2. a description of a solar-driven instrumental Al Kα component seen in nightside CLASS spectra when the spacecraft

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is sunlit, and its treatment in the background model;

3. an error budget from synthetic recovery tests, including the effect of errors in the solar temperature, which indicates which element ratios can be recovered with this approach;
4. global Mg/Si and Al/Si maps from 24 monthly archives, with observation-count and uncertainty layers, compared with returned-sample compositions at nine landing sites and with Lunar Prospector GRS maps, using analysis choices fixed before the comparison.

We report element ratios relative to Si, not absolute abundances, and we state which quantities the data do not constrain with this approach. The maps are intended to complement, not replace, the XSM-based analyses of the instrument team.

2. Data

2.1. CLASS Level-1 spectra

The Chandrayaan-2 Large Area Soft X-ray Spectrometer (CLASS) carries sixteen e2v CCD-236 swept-charge devices (SCDs) of 4 cm^2 each, behind a collimator with a field of view of about 14° full width (Narendranath et al., 2014; Radhakrishna et al., 2020; Pillai et al., 2021). We use the publicly released Level-1 product from the PRADAN portal of the Indian Space Science Data Centre. Each Level-1 file is an OGIP pulse-height table of 2048 channels holding the counts summed over the SCDs listed in the `SCD_USED` keyword for one 8 s integration, together with the four footprint corner coordinates, the solar incidence angle `SOLARANG`, the emission angle `EMISNANG`, the exposure time, detector temperature and the processing-software version.

We downloaded every monthly archive available to us: 24 archives (December 2020; January and February 2021; March 2022; March 2024; and every month from November 2024 to May 2026), totalling 1,515,239 Level-1 exposures. The archives are non-contiguous because CLASS observes only when the spacecraft attitude, power and data-volume budgets allow, and because releases on PRADAN lag the observations. The 2020–2022 months fall near the minimum of solar cycle 25 (Hathaway, 2015) and contribute few usable spectra (Section 7.1).

A survey of 4,800 headers sampled uniformly from all months (Table 1) shows that every file has 2048 channels and a nominal gain keyword of 13.5 eV/channel. No 1024-channel products occur in this release, so the 1024-channel branch of earlier processing is not used. The emission angle is $< 3{\times}10^{-6}$°, so the instrument views at nadir. Exposures are 8 s except for 0.6% partial integrations (1.8 s to 7.8 s). The number of SCDs summed varies: 26% of exposures use fewer than 16 devices, so the effective area changes from file to file. No response matrix (`RESPFILE`), ancillary response (`ANCRFILE`) or efficiency information ships with the release, and the background keyword is `NONE`.

2.2. GOES XRS solar soft X-ray data

Within the CLASS archive itself there is no measurement of the incident solar spectrum. We characterise the incident solar spectrum using the GOES X-Ray Sensor (XRS) 1-minute science-quality averages from NOAA/NCEI (GOES-16 for 2020–2022; GOES-18 for 2024–2026). The Moon and the Earth see the same solar disk to within a fraction of a degree. For each minute we derive the isothermal temperature T and emission measure from the ratio of the 0.05 nm to 0.4 nm and 0.1 nm to 0.8 nm channels (White et al., 2005), using the CHIANTI-based response tables with coronal abundances implemented in `sunkit-instruments` (The SunPy Community et al., 2020). Each CLASS exposure is assigned the values of the nearest GOES minute to its mid-time (tolerance 1 min).

Below the B1 flux level ($F_{0.1-0.8\,\mathrm{nm}} < 1 \times 10^{-7}\,\mathrm{W\,m^{-2}}$), the GOES-16 ratio temperature is unreliable: it is either pinned at the lower edge of the response table (2.8 MK) or diverges (Section 9). Exposures taken below B1 are therefore excluded from all maps. This threshold was fixed before any comparison with ground truth.

2.3. Calibration and validation data

For validation we use the best-estimate average soil compositions of the Apollo 11, 12, 14, 15, 16 and 17 and Luna 16, 20 and 24 landing sites, compiled from returned-sample analyses (Heiken et al., 1991, Table 7.15; Basu and Riegsecker, 1998) (Table 4), and the Lunar Prospector gamma-ray spectrometer (GRS) oxide maps of Prettyman et al. (2006). The latter are distributed by the PDS Geosciences Node on 5° equal-area pixels (Prettyman, 2012). Oxide weight fractions are converted to element weight ratios relative to Si using standard stoichiometric factors.

3. Forward model

We model the lunar regolith within one footprint as a flat, homogeneous, infinitely thick target. Its composition is parameterised by element-to-silicon weight ratios $r_i = C_i/C_{\mathrm{Si}}$, and the observed spectrum is the sum of four components: fluorescence lines, scattered solar continuum and lines, particle background, and a solar-driven instrumental component. All atomic parameters come from xraylib 4.2 (Schoonjans et al., 2011).

3.1. Composition closure

The fitted elements are Mg, Al, Ca, Ti and Fe, each expressed relative to Si. Each element is assumed to be present as its common lunar oxide (SiO_2, MgO, Al_2O_3, CaO, TiO_2, FeO). Minor oxides are fixed at typical soil values (Na_2O 0.5, K_2O 0.1, MnO 0.15, Cr_2O_3 0.2 wt%). Requiring the oxides to sum to unity fixes the absolute Si weight fraction:

$$C_{\mathrm{Si}} = \frac{1 - \sum_m w_m^{\mathrm{ox}}}{f_{\mathrm{Si}} + \sum_i r_i f_i}, \tag{1}$$

where f_i is the oxide-to-element mass factor and w_m^{ox} are the fixed minor-oxide fractions. Oxygen follows stoichiometrically, so it is not a free parameter. The matrix mass attenuation coefficient is $\mu(E) = \sum_j C_j \mu_j(E)$, with μ_j the total mass attenuation coefficient of element j.

Table 1: CLASS Level-1 header survey (4,800 files, 200 per month).

Keyword	Value / range
`DETCHANS`	2048 in all files
`GAIN`	13.5 eV/channel in all files (fitted in-flight value: Section 3.6)
`EXPOSURE`	8 s (99.4%); partial integrations 1.8 s to 7.8 s
`SCD_USED`	16 SCDs in 74%; 1–15 otherwise
`EMISNANG`	$< 3 \times 10^{-6}$° (nadir)
`SOLARANG`	0° to 166° (= `PHASEANG`)
`TEMP`	−54 °C to −22 °C (median −41 °C)
`SW_VERSN`	2.0 (Dec 2020–Feb 2021), 2.1 (later)
`RESPFILE`, `ANCRFILE`	empty

3.2. Incident solar spectrum

The incident photon spectrum $S(E;T)$ is the isothermal CHIANTI spectrum, lines plus continuum with coronal abundances (Dere et al., 1997), evaluated with `sunkit-spex` at the GOES temperature of each exposure (Section 2.2). It is tabulated on a 5 eV grid from 1.01 keV to 20 keV at 61 logarithmically spaced temperatures between 2 and 40 MK, and interpolated in $\log T$. Its absolute normalisation is absorbed by a free amplitude.

Line emission dominates the excitation of the light elements. Between 3 and 15 MK, 63–80% of the photon flux above the Mg K edge, and 55–74% above the Si K edge, is in lines, mainly the He-like Mg XI, Si XIII and S XV complexes. A bremsstrahlung-only spectrum would therefore misrepresent the relative excitation of Mg, Al and Si (Section 9).

3.3. Primary fluorescence

For line ℓ (Kα or Kβ) of element i, the primary fluorescence intensity follows Sherman (1955):

$$P_{i\ell} = C_i\,\omega_{i,K}\,p_{i\ell}\int_{E_{K,i}}^{E_{\max}} \frac{S(E;T)\,\tau_{i,K}(E)}{\mu(E)/\sin\psi_1 + \mu(E_{i\ell})/\sin\psi_2}\,\mathrm{d}E, \quad (2)$$

where $\tau_{i,K}$ is the K-shell photoionisation cross-section, $\omega_{i,K}$ is the K-shell fluorescence yield, $p_{i\ell}$ is the radiative rate of the line, and $E_{K,i}$ is the K-edge energy. The incidence elevation is $\psi_1 = 90° - $ `SOLARANG`, and the emergence elevation is $\psi_2 = 90°$ (nadir). The product $\omega_{i,K}p_{i\ell}\tau_{i,K}$ reproduces the xraylib fluorescence-line cross-section to within 5%. The integral is evaluated on the 5 eV grid up to $E_{\max} = 15$ keV; a unit test shows this agrees with adaptive quadrature to within 1% and with a 1 eV grid to within 0.3%. Because the integrand depends on the composition through μ, Eq. (2) couples all elements. This coupling is the matrix effect; ratios of line intensities are therefore not equal to ratios of concentrations.

3.4. Secondary fluorescence

Fluorescence of element i excited by line m of element j ($E_{jm} > E_{K,i}$) is computed with the thick-target expression of de Boer (1990), which is linear in both C_i and C_j. Over 3 MK to 15 MK and both mare and highland compositions, the secondary contribution to every Kα line is at most 1.3% of the primary. We therefore omit it from the fitted model. We do, however, include it when generating the synthetic spectra of Section 5, so the bias from omitting it is part of the measured recovery bias.

3.5. Scattered solar spectrum

Coherent and incoherent scattering of the solar spectrum by the regolith contributes a continuum, and at the GOES temperatures encountered, scattered solar lines as well:

$$\Sigma(E) = S(E;T)\,\frac{\sigma_s(E)}{\mu(E)/\sin\psi_1 + \mu(E)/\sin\psi_2}, \quad (3)$$

where $\sigma_s = \sum_j C_j(\sigma_{\mathrm{coh},j} + \sigma_{\mathrm{inc},j})$. The angular dependence of the differential scattering cross-sections is not modelled. Instead, the scattered component carries its own free amplitude B.

3.6. Detector response and energy scale

The centre energy of channel k is $E_k = g\,k + o$. Each line is distributed over channels by a Gaussian integrated over the channel boundaries $E_k^{\pm} = E_k \pm g/2$:

$$R_{\ell k} = \frac{\Phi\!\left(\frac{E_k^{+}-E_\ell}{\sigma_\ell}\right) - \Phi\!\left(\frac{E_k^{-}-E_\ell}{\sigma_\ell}\right)}{\mathrm{erf}(5/\sqrt{2})}, \qquad \sigma_\ell^2 = \sigma_0^2 + F\,w\,E_\ell, \quad (4)$$

where Φ is the standard normal cumulative distribution, truncated at $\pm 5\sigma$, $F = 0.115$ is the Fano factor of silicon and $w = 3.68$ eV. The normalisation conserves line area to 10^{-9} (unit test). The scattered spectrum, Eq. (3), is folded through the same response on the 5 eV grid.

The gain g, offset o and noise term σ_0 are determined for each month. We stack all 16-SCD dayside exposures of the month, subtract the particle background, and fit the Mg, Al, Si and Ca K lines jointly. For example, for December 2024 (75,707 exposures) we find $g = 13.74$ eV/channel, $o = 2.2$ eV and $\sigma_0 = 74$ eV, i.e. 185 eV FWHM at Al Kα. The fitted gain is about 1.6% larger than the header value of 13.5 eV/channel. Using the header gain shifts the Si Kα centroid by about two channels. The per-month values are listed in Table A.6.

Each SCD views the surface through two optical-blocking filters, one on either side of the collimator, each made of 0.2 µm polyimide coated with 0.2 µm aluminium (Radhakrishna et al., 2020; Pillai et al., 2021). We apply their X-ray transmission at the photon energy, before folding. Because the Al K edge (1.56 keV) lies between the Al and Si lines, the transmission is 0.91 for Al Kα but 0.69 for Si Kα (Mg Kα: 0.86). Ignoring the filters would inflate Al/Si by 31% and Mg/Si by 24%. The

quantum efficiency of the SCDs themselves (depletion depth, surface dead layer) is not published and is not modelled (Section 9.5).

3.7. Background: particles and a solar-driven instrumental component

Our earlier processing of these data subtracted the mean nightside spectrum (SOLARANG $\geq 90°$). This mean appears to contain two physically distinct components (Fig. 1).

The first is the particle background, a smooth continuum with weak features. It dominates exposures taken while the spacecraft is in the lunar umbra, i.e. when SOLARANG exceeds $90° + \arccos[R_M/(R_M + h)] \approx 109°$ at $h = 100$ km. The second is a strong Al Kα line, together with a soft continuum and a feature near 6.7 keV. This second component appears only while the spacecraft itself is sunlit. In December 2024 the median background-corrected Al Kα rate of nightside exposures is 31.8 counts s^{-1} when the spacecraft is sunlit (n = 8,555), against 0.16 counts s^{-1} in the umbra (n = 2,254). Its rate correlates with solar proxies (Spearman $\rho = 0.75$ with the 0.6 keV to 0.9 keV counts) but not with the 8 keV to 20 keV particle rate ($\rho = 0.06$), and it varies on time scales shorter than 30 min. We interpret it as solar X-ray fluorescence of aluminium in the instrument structure. If present on the dayside at the nightside level, it would be comparable in magnitude to the lunar Al line.

We therefore build two templates per month, in units of counts s^{-1} per 16 SCDs. The first is $b_k^{\rm p}$, the mean umbra spectrum. The second is $b_k^{\rm s}$, the mean sunlit-nightside spectrum minus $b_k^{\rm p}$ scaled to the same 8 keV to 20 keV rate. Months with fewer than 100 umbra (or sunlit) nightside exposures use the December 2024 template shape. We could not predict the instrumental amplitude from GOES with useful accuracy: the GOES-predicted Al-exciting flux explains it only to a factor of $\approx$2 (Spearman $\rho = 0.39$), with an angle-dependent scale.

The geometry of Fig. 1 suggests that the instrumental component requires the Sun to be behind the spacecraft: it is already small at SOLARANG $\approx$ 90–93°, and we therefore expect it to be negligible on the dayside, where the Sun is in front of the nadir-pointing instrument. Our primary model therefore fixes its amplitude to $\gamma = 0$ for all dayside spectra, so the background is $\beta\, b_k^{\rm p}$. We also ran an alternative model in which γ is a free, non-negative parameter. It is reported in Section 8.4, together with the reason the primary model was chosen.

4. Spectral fitting

4.1. Likelihood

Each exposure is fitted on its raw counts d_k without smoothing, denoising or background subtraction. The fitted channels are those with $E_k \in [1.15, 7.5]$ keV. The lower limit avoids the Na Kα line at 1.04 keV and a steep excess below 1.1 keV that is present in every stacked spectrum. One additional bin holds the summed counts in 8 keV to 20 keV, where no lunar fluorescence is expected. That bin fixes the particle-background level without depending on the background shape. The expected counts are

$$m_k = A\Big[\sum_\ell \frac{P_\ell}{P_{\rm ref}} R_{\ell k} + B\,\frac{(\mathcal{F}\Sigma)_k}{\Sigma_{\rm ref}}\Big] + \beta\, b_k^{\rm p}\, t\, \frac{n_{\rm SCD}}{16}, \qquad (5)$$

where t is the exposure, $n_{\rm SCD}$ is the number of SCDs summed, $\mathcal{F}$ is the detector folding of Eq. (4), and $P_{\rm ref}$, $\Sigma_{\rm ref}$ are fixed reference normalisations. The parameter vector is $\theta = (A, r_{\rm Mg}, r_{\rm Al}, r_{\rm Ca}, B, \beta)$. In the alternative model of Section 8.4, the term $\gamma\, b_k^{\rm s}\, t\, n_{\rm SCD}/16$ is added with $\gamma \geq 0$ free. All components are non-negative, the ratios are bounded to $[0, 5]$, and $\beta \leq 10$.

Si is the reference element, so only ratios and one amplitude are fitted. This avoids the scale degeneracy of fitting absolute concentrations together with a free global scale. Ti/Si and Fe/Si are held at global lunar-soil means (0.04 and 0.37) and enter only through the matrix. At GOES temperatures of 3–8 MK the solar flux above the Ti and Fe K edges is too small for an 8 s spectrum to constrain them; leaving them free made 92% of test fits ill-conditioned. We minimise the Cash statistic (Cash, 1979)

$$C = 2\sum_k \left[m_k - d_k + d_k \ln(d_k/m_k)\right] \qquad (6)$$

with a trust-region reflective least-squares solver applied to the signed deviance residuals, whose sum of squares equals C.

4.2. Uncertainties, goodness of fit and quality flags

Parameter covariances are taken from the inverse Poisson Fisher matrix $\mathcal{I} = J^\top \mathrm{diag}(1/m) J$, where $J = \partial m/\partial\theta$ is computed by finite differences at the optimum. We also record the correlation matrix and the condition number of its core block $(A, r_{\rm Mg}, r_{\rm Al}, B, \beta)$. For goodness of fit we compare C over the main window with its expectation and variance for the fitted model, computed exactly for Poisson counts (Kaastra, 2017): $z = (C - \langle C\rangle)/\sqrt{\mathrm{Var}\, C}$.

A spectrum enters the maps (quality = 0) only if all of the following hold:

- the fit converged;
- there are at least 300 counts in the 1.15 keV to 7.5 keV window;
- $|z| \leq 3$;
- neither Mg/Si nor Al/Si is at a bound;
- the core condition number is below 10^4;
- the geometry is valid ($0.5° < \psi_1 \leq 90°$, altitude > 0);
- a reliable GOES temperature exists (flux $\geq$ B1).

These criteria, the 2° mapping grid and the validation radii were fixed and logged before any comparison with reference data. The count threshold is varied (300, 1000, 3000) as a sensitivity test. No flux-percentile selection is applied.

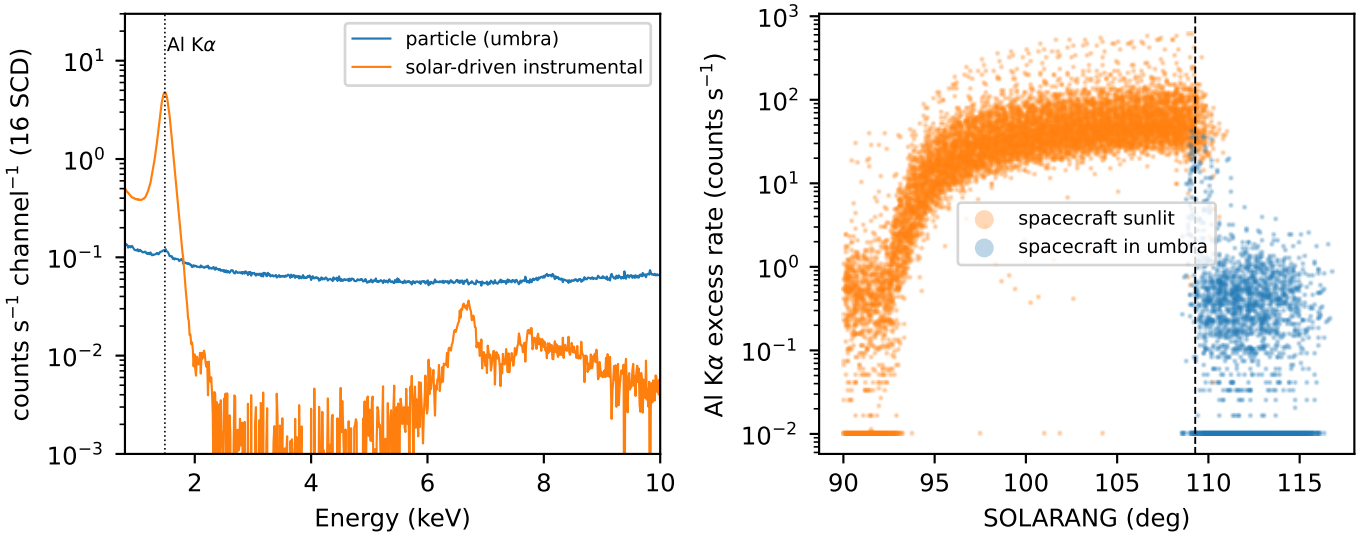


Figure 1: Background components of CLASS spectra. Left: December 2024 particle-background template (spacecraft in the lunar umbra) and solar-driven instrumental template (sunlit nightside minus particle background). Right: Al Kα excess rate of nightside exposures against SOLARANG; the dashed line is the shadow boundary of the spacecraft.

4.3. Products

Ca/Si is fitted because it shapes the model between the Si and Ca lines, but it is not reported. The synthetic tests of Section 5 show it is biased even at the correct temperature and changes by an order of magnitude under realistic temperature errors. The products are Mg/Si and Al/Si weight ratios with per-footprint uncertainties.

5. Synthetic recovery tests

Before applying the fitter to real data, we tested it on synthetic CLASS-like spectra.

Generator. The synthetic spectra are generated with the full forward model, including secondary fluorescence, which the fitted model omits. The calibration is the December 2024 one, the background uses the real December 2024 particle template ($\beta = 1$), and each spectrum is an 8 s, 16-SCD exposure with Poisson noise. Compositions are drawn uniformly from lunar ranges (Mg/Si 0.10–0.35, Al/Si 0.20–0.80, Ca/Si 0.30–0.60), with incidence angles of 10–80°. The scattered-solar amplitude is set to match the continuum level of the December 2024 stack.

Results at the correct temperature. The signal level is expressed as the number of Si Kα counts. For a 5 MK solar spectrum (the median GOES temperature of 2024–2026), Table 2 lists the median relative bias and robust (MAD-based) relative scatter of the recovered ratios. It also lists the width of the pull distribution, $(\hat{r} - r)/\hat{\sigma}_r$. Mg/Si and Al/Si are recovered with $|\text{bias}| \leq 4\%$ for every signal level, and with pulls of width 0.84–1.25, which indicates that the Fisher uncertainties are approximately correct. Omitting secondary fluorescence does not introduce a noticeable bias. In the alternative model, where the generator adds the instrumental template ($\gamma = 0.5$) and the fit leaves γ free, the recovery is equally unbiased, but the Al/Si scatter at 300 counts roughly doubles. Ca/Si is biased by up to +50%, with 50–150% scatter, even when the temperature is exact, because a 5 MK spectrum barely excites Ca.

Temperature error. The main systematic is the solar temperature. We generate spectra at one temperature and fit them assuming another, to mimic error in the GOES temperature. With a true spectrum 30% hotter than assumed, Mg/Si is biased by −15 to −13% and Al/Si by +3 to +5%. With a true spectrum 20% cooler, the biases are +28 to +33% and +5 to +7%. Ca/Si changes by a factor of ∼9 or collapses to zero. These numbers set the systematic uncertainty quoted for the maps (Section 9.5), and they are the reason we do not report Ca/Si, Ti/Si or Fe/Si with this approach.

6. Mapping

Gridding. Footprints are placed on a 2° latitude–longitude grid. Each footprint quadrilateral is sampled with a 5×5 bilinear grid of points between its four corners, and its weight is shared equally among the grid cells containing those points. The footprint corners recorded in the Level-1 headers span a median of 25 km across track by 37 km along track. This is the collimator field of view, about 25 km at 100 km altitude, plus roughly 13 km of ground-track motion during the 8 s integration. One degree is 30.3 km at the lunar equator, so a footprint overlaps one to four 2° cells.

Cell values and quality layers. Each cell value is the inverse-variance-weighted mean of the footprint ratios. Three quality layers accompany each map:

- the number of footprints contributing to the cell;

Table 2: Synthetic recovery at a 5 MK solar temperature (80 spectra per level): median relative bias / robust relative scatter (%), and pull width.

Si Kα counts	300	1,000	3,000	10,000
Mg/Si bias / scatter (%)	+1 / 21	−4 / 10	−1 / 7	−1 / 5
Mg/Si pull width	0.84	0.86	1.13	1.25
Al/Si bias / scatter (%)	+2 / 16	0 / 8	−1 / 4	0 / 3
Al/Si pull width	0.91	0.97	0.93	1.14
Ca/Si bias / scatter (%)	+65 / 245	+38 / 127	+52 / 73	+27 / 55
Ca/Si pull width	0.87	0.90	1.08	1.82
Quality-0 fraction	1.00	1.00	1.00	1.00

Table 3: Exposure counts at each pipeline stage (all 24 months).

Stage	Exposures
Level-1 files	1,515,239
Dayside, valid geometry (all fitted)	1,053,989
with reliable GOES temperature (≥ B1)	1,017,226
fit converged	1,037,846
Mapping set (quality = 0, ≥300 counts, ≥ B1)	578,530

- the standard error of the weighted mean;
- the weighted scatter of the footprint ratios.

Coverage. We report coverage separately for all dayside footprints and for the quality-0 mapping set. We give it both as a fraction of grid cells and as a fraction of lunar surface area, where cells are weighted by cos(latitude). No smoothing, interpolation or infill enters any statistic, map value or validation number in this paper.

Resolution. The effective spatial resolution of a single measurement is set by the footprint. Its full extent is about 25 × 37 km, and the collimator response has a narrower FWHM, quoted as 12.5 km by Pillai et al. (2021). The 2° cells (61 km at the equator) set the resolution of the maps. The grid spacing should not be read as a resolution finer than either of these.

7. Results

7.1. Pipeline statistics

Table 3 traces the exposures through the pipeline. All 1,053,989 dayside exposures with valid geometry were fitted; no flux-percentile pre-selection was applied. 578,530 spectra (54.9% of the dayside set) pass every quality criterion and form the mapping set. The fraction passing varies strongly with solar activity. It ranges from 2% in February 2021, near solar minimum, where most spectra fall below the count threshold or below GOES B1, to 66% in February 2025 (Table A.6). The GOES temperatures of the mapped exposures have a median of 5.2 MK (5th–95th percentile 3.6–8.0 MK).

7.2. Fit examples

Figure 2 shows fits to a bright, a median and a faint dayside spectrum. The model describes the Mg, Al and Si K lines, the Ca K line and the scattered solar continuum and lines. In the faint spectrum the particle background dominates above 2 keV, and the uncertainties on Mg/Si and Al/Si become correspondingly large. Bright flare spectra (e.g. the top panel, T = 16.8 MK) often fail the goodness-of-fit criterion. Residual structure below 2 keV points to limitations of the isothermal solar spectrum and of the line-shape model, so the mapping set is weighted towards non-flaring and moderate-flare conditions.

7.3. Global maps

Figure 3 shows the Mg/Si and Al/Si maps with their observation-count and standard-error layers. Over the mapping set, the median footprint ratios are Mg/Si = 0.22 (5th–95th percentile 0.07–0.43) and Al/Si = 0.92 (0.49–1.38). The Mg/Si range lies within the range of returned lunar soils. The Al/Si median is higher than most returned soils, which may indicate a residual efficiency or calibration offset between the Al and Si lines (Section 9.5). An offset of this kind would scale all values without changing the spatial pattern or the correlations, so we regard the maps as relative.

Quality-0 footprints reach every 2° cell (100.00% of cells, 100.0% of the area), and 99.7% of the surface is covered by cells with at least five footprints. The coverage comes from the polar orbits of the 24 archives at all local times, not from interpolation. The polar cells have the fewest footprints and the largest standard errors.

The Mg/Si map shows the expected first-order dichotomy: higher values over the nearside maria and the South Pole–Aitken basin, lower values over the farside feldspathic highlands. The Al/Si map shows the complementary pattern at higher contrast. Several maria (Imbrium, Serenitatis, Crisium, Tranquillitatis, Fecunditatis, Nubium, Humorum) and the South Pole–Aitken basin appear as low-Al/Si units against the feldspathic highlands. Residual north–south striping follows individual orbits. We attribute it mainly to orbit-to-orbit differences in the solar state that the GOES isothermal temperature does not fully capture (Section 9).

8. Validation

The following choices were fixed and logged before the validation of the primary model was run:

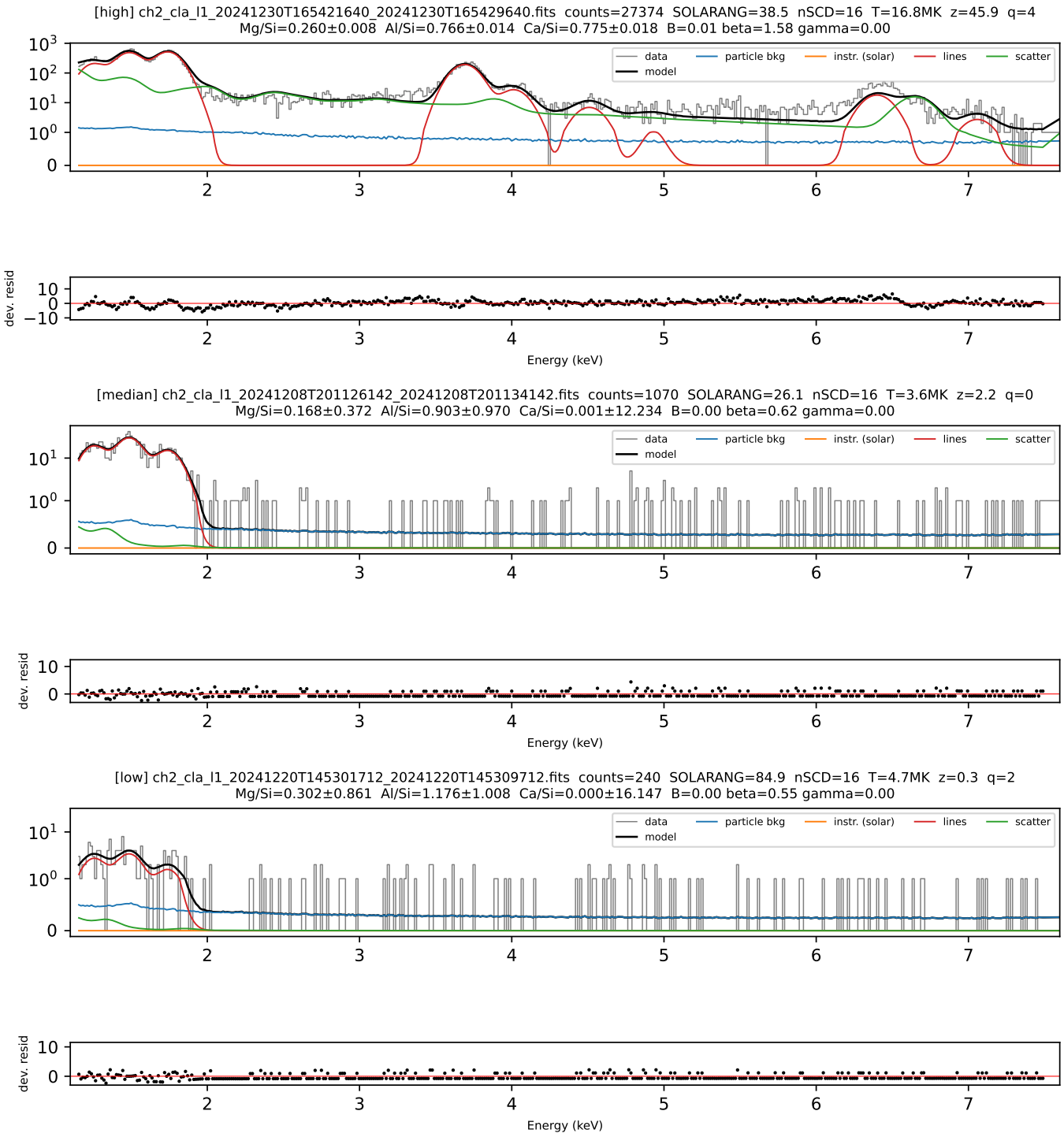


Figure 2: Example spectral fits. Fits (December 2024) to a bright, a median and a faint dayside spectrum: data (grey), total model (black), fluorescence lines (red), scattered solar spectrum (green), and particle background (blue), with deviance residuals.

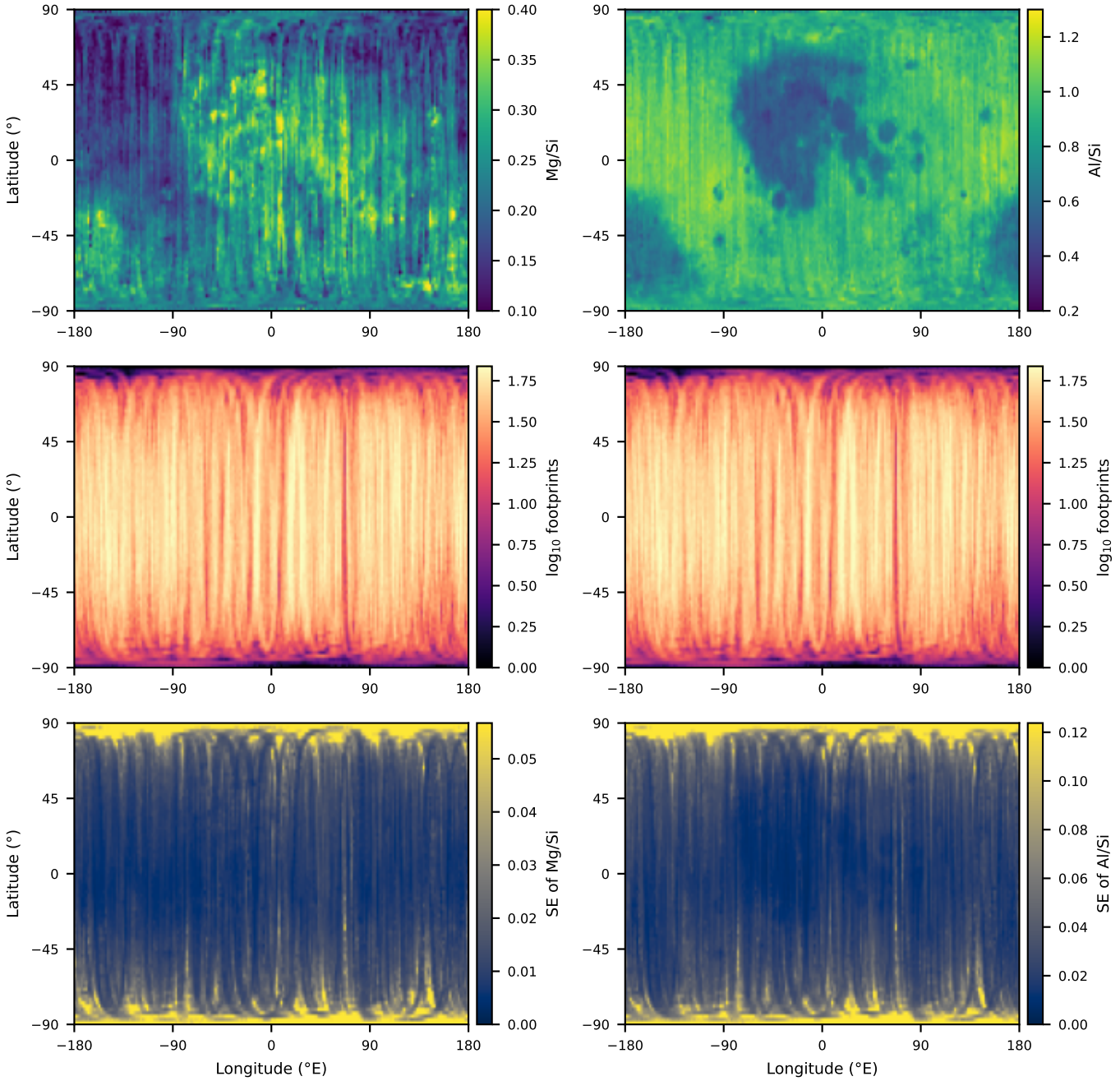


Figure 3: Global Mg/Si and Al/Si maps with quality layers. Mg/Si (left) and Al/Si (right) weight ratios on a 2° grid (inverse-variance-weighted means of quality-0 footprints; no smoothing or infill), with the number of footprints per cell (middle) and the standard error of the cell mean (bottom).

- the matching radii (1, 2 and 5°, with 2° primary);
- the statistics: Pearson and Spearman coefficients with bootstrap 95% confidence intervals, leave-one-site-out ranges, and a 20° spatial block bootstrap for GRS;
- the criterion for regarding an element as supported by both comparisons: Pearson $r > 0$ with a 95% confidence interval excluding zero in both the landing-site and GRS tests.

The history of the primary model, including an earlier model whose validation motivated it, is given in Section 8.4.

8.1. Landing sites

Table 4 lists the returned-soil ratios (average soil compositions; Heiken et al., 1991, Table 7.15; Basu and Riegsecker, 1998) and the CLASS weighted means within 2° of each site.

- *Al/Si, 2°:* $r = 0.81$ (95% CI [0.48, 0.97]), Spearman $\rho = 0.78$ ($p = 0.01$), leave-one-out range 0.81–0.86.
- *Al/Si, other radii:* $r = 0.71$ (CI [0.51, 0.99]) at 1° and $r = 0.47$ at 5°.
- *Mg/Si, 2°:* $r = 0.38$ ([−0.50, 0.83]); $r = -0.64$ at 1° and 0.09 at 5°.

The CLASS site means have small statistical errors (Table 4). Their scatter about the soil values is therefore mostly systematic. Footprints of 25 × 37 km average over terrain much larger than the sampled landing sites, and several sites lie on mare–highland contacts (Apollo 15, 16, 17; Luna 20). Al/Si meets the site criterion; Mg/Si does not.

8.2. Lunar Prospector GRS

All 1,790 5° equal-area GRS pixels contain at least 20 CLASS footprints. On these pixels:

- *Al/Si:* $r = 0.72$ (block-bootstrap 95% CI [0.66, 0.76]; Spearman $\rho = 0.60$);
- *Mg/Si:* $r = 0.61$ ([0.56, 0.66]; $\rho = 0.63$).

Both meet the GRS criterion (Fig. 4). The correlations are stable under the count-threshold test: Al/Si $r = 0.72$, 0.72 and 0.69, and Mg/Si $r = 0.61$, 0.58 and 0.58, for thresholds of 300, 1000 and 3000 counts (578,530, 143,961 and 11,259 spectra). Restricting to GOES ≥ B3 leaves them unchanged.

8.3. Interpretation of the correlation values

Agreement with GRS is bounded by noise in both data sets. We split the mapping set into two halves by alternate months, map each half separately, and correlate the two maps. After Spearman–Brown correction, the CLASS pixel means have reliabilities of 0.97 (Al/Si) and 0.76 (Mg/Si). The GRS error matrices imply reliabilities of 0.78 and 0.73. The maximum attainable correlations are therefore $\sqrt{\rho_{\mathrm{CLASS}}\rho_{\mathrm{GRS}}} = 0.87$ for Al/Si and 0.75 for Mg/Si. The observed values are about 80% of these ceilings. The GRS Mg and Al abundances are themselves model-dependent (Prettyman et al., 2006).

The landing-site test has little discriminating power at this scale. Applied to the same nine soil averages, using the 5° pixel containing each site, GRS itself gives $r = -0.34$ for Al/Si and -0.11 for Mg/Si. Soil averages sample metres to kilometres around each landing point, whereas orbital pixels average tens to hundreds of kilometres (Basu and Riegsecker, 1998). With nine sites and this mismatch in scale, the site test has limited power, and its outcome for Mg/Si should be interpreted with caution.

Comparison with the earlier processing. The earlier processing of the same data (centre-point binning of denoised least-squares fits without a solar model) gives $r = 0.19$ (Al/Si) and 0.48 (Mg/Si) on the same GRS pixels.

Weighted median. Replacing the weighted mean by a weighted median changes little: GRS $r = 0.71$ (Al/Si) and 0.58 (Mg/Si). Figure 5 shows the weighted-median maps.

8.4. Alternative model and the history of the primary model

The analysis was first run with the instrumental amplitude γ free (alternative model, 429,217 mapped spectra). Under the same pre-declared rules, that model gave the following.

- *GRS:* Al/Si $r = 0.35$ ([0.26, 0.44]) and Mg/Si $r = 0.60$ ([0.54, 0.65]).
- *Landing sites (2°):* Al/Si $r = 0.43$ and Mg/Si 0.37. Neither element was claimed under the criterion.
- *Diagnosis:* the Al/Si reliability was only 0.42 (ceiling 0.57). In a fixed farside region, the monthly median Al/Si anticorrelated with the fraction of spectra having a non-zero fitted γ (Spearman −0.88): in some months the free γ absorbed part of the lunar Al line.
- *Post-hoc check:* restricting that model to spectra with $\gamma <$ 0.05 (51%) gave GRS Al/Si $r = 0.72$.

Because the nightside geometry (Fig. 1) suggests, independently of the reference data, that the component is negligible on the dayside, we then specified the primary model ($\gamma \equiv 0$ on the dayside). We logged it, together with the commitment to report its outcome whatever it was, *before* running it on the full data set. All other choices are identical to the alternative model.

During development we also corrected one defect after a first validation. The boundary threshold used to flag pinned ratios was raised from 10^{-6} to 10^{-3}, because 254 fits pinned near zero had collapsed Fisher uncertainties and dominated the inverse-variance weights. Before that fix, the alternative model gave GRS Al/Si $r = 0.21$.

The primary model was thus motivated by the diagnosis of the alternative model's validation, and we note this so that readers can weigh it accordingly. Its specification was fixed before it was run, and it rests on a physical argument that does not involve the reference data.

8.5. Outcome

Under the pre-declared rule, Al/Si meets both criteria: GRS ($r = 0.72$) and the landing sites ($r = 0.81$). Mg/Si meets the

Table 4: Landing-site validation at 2° radius: returned-soil weight ratios and CLASS weighted means ± standard error.

Site	Lat	Lon	Mg/Si		Al/Si	
	(°)	(°E)	soil	CLASS	soil	CLASS
Apollo 11	0.67	23.47	0.238	0.311±0.005	0.365	0.679±0.008
Apollo 12	-3.01	-23.42	0.259	0.282±0.005	0.316	0.545±0.007
Apollo 14	-3.65	-17.47	0.252	0.324±0.003	0.410	0.639±0.005
Apollo 15	26.13	3.63	0.317	0.322±0.007	0.353	0.643±0.010
Apollo 16	8.97	15.50	0.163	0.292±0.006	0.687	0.758±0.010
Apollo 17	20.19	30.77	0.311	0.302±0.006	0.448	0.672±0.009
Luna 16	-0.68	56.30	0.272	0.314±0.004	0.415	0.662±0.006
Luna 20	3.53	56.55	0.280	0.300±0.004	0.560	0.811±0.008
Luna 24	12.71	62.20	0.276	0.308±0.004	0.322	0.649±0.006

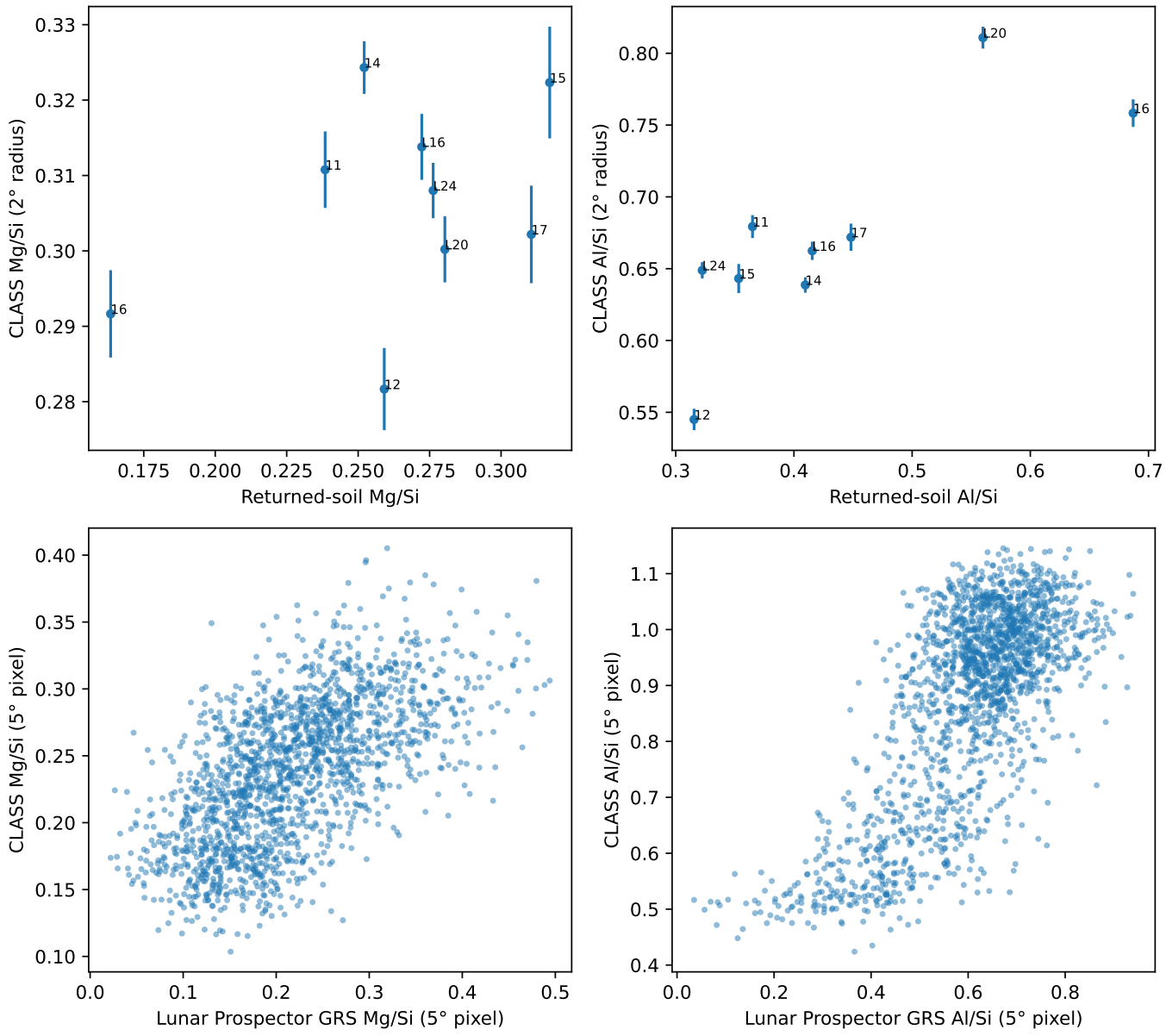


Figure 4: Comparison with landing-site soils and Lunar Prospector GRS. Top: CLASS ratios within 2° of the landing sites against returned-soil ratios. Bottom: CLASS against Lunar Prospector GRS ratios on 5° equal-area pixels.

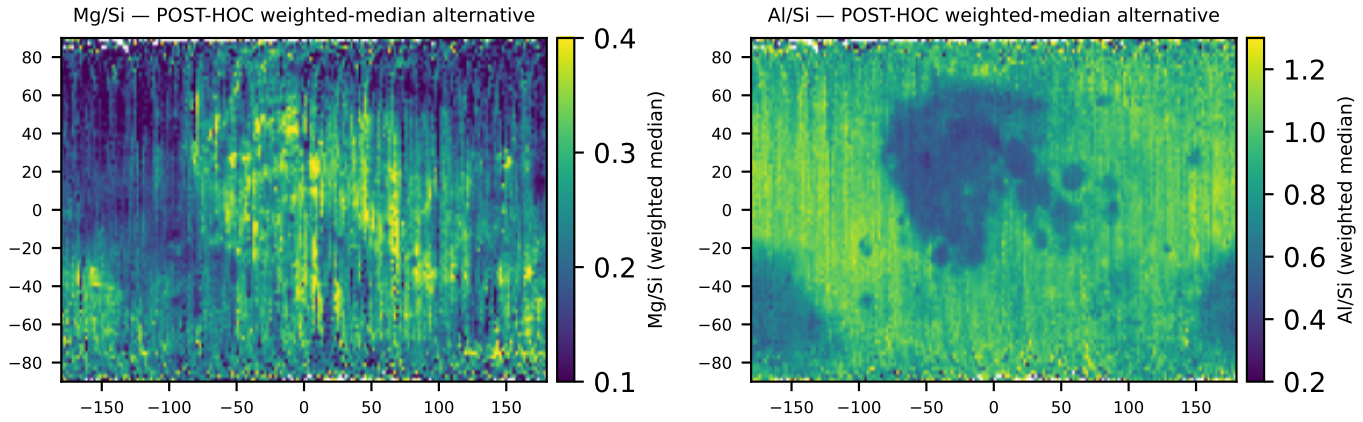


Figure 5: Weighted-median maps. Alternative cell statistic (not the primary product): inverse-variance-weighted median of footprint ratios per 2° cell (footprint centres).

GRS criterion ($r = 0.61$) but not the nine-site criterion, which GRS itself also does not meet. We therefore present Al/Si as a relative map that is supported by both comparisons, and Mg/Si as a relative map whose large-scale pattern agrees with GRS, without a claim of site-level accuracy.

9. Discussion

9.1. Sensitivity of line ratios to the solar spectrum

Element/Si ratios are sometimes expected to be insensitive to the incident spectrum. Our forward model indicates that the sensitivity is appreciable. Each K line is excited by the part of the solar spectrum above its own edge (Mg 1.31 keV, Si 1.84 keV, Ca 4.04 keV), and at coronal temperatures that part is dominated by a few He-like line complexes whose relative strengths change steeply with temperature.

In our synthetic tests, a spectrum 30% hotter than assumed biases Mg/Si by −15 to −13% and changes Ca/Si by an order of magnitude. Refitting real data with the GOES temperature scaled by 0.8 and 1.2 changes the median Mg/Si by −25% and +27%, and Al/Si by −9% and +8%.

With a continuum-only reference spectrum, we could not find a single temperature that gave lunar-like Mg/Si and Ca/Si at the same time (Mg/Si 0.3–0.8 with Ca/Si 0.004–0.5). The implausible ratios of our own earlier processing of these data (Mg/Si ≈ 1.4, Ca/Si ≈ 0.01) probably have the same origin. For this reason we consider a time-resolved description of the solar spectrum important even for ratio maps.

9.2. GOES as a proxy for the solar spectrum

GOES provides an estimate of the solar state for every CLASS exposure from public data, with known limitations. The two-channel temperature is a single isothermal value, weighted towards the hottest plasma, and below the B1 level it is unreliable; in our data it was pinned to the edge of the response table or diverged there. The GOES-16 and GOES-18 temperatures agree to 0.5% in their overlap.

The real corona is multi-thermal, and the isothermal approximation underpredicts the flux above 4 keV at the temperatures typical of our data. This is a likely reason why Ca/Si is biased even in self-consistent synthetic tests and why Ti and Fe are not constrained. The residual orbit-to-orbit striping in the maps is consistent with solar-state errors of this kind.

9.3. Comparison with XSM solar spectra

To test the GOES approximation directly, we used Level-2 spectra of the Chandrayaan-2 Solar X-ray Monitor (Mithun et al., 2020) for three months chosen in advance (December 2024, June 2025 and December 2025). XSM views the Sun for part of each orbit, and its 1-minute spectra overlap 6,327 mapping-set CLASS exposures (1,386 XSM minutes). The test and its acceptance criterion were fixed before it was run. We fitted each XSM minute in 1.5 keV to 10 keV with the same CHIANTI table, using the response matrix of the XSM calibration database, first with a single temperature and then with two temperatures. We then refitted the overlapping CLASS spectra with the XSM-derived spectrum in place of the GOES one, keeping everything else unchanged. Refitting with the GOES temperature reproduces the mapped values exactly.

The XSM isothermal temperatures are higher than the GOES values by a median factor of 1.12 (16th–84th percentile 1.01–1.17). With the XSM isothermal spectrum, the median Al/Si changes by +4% (16th–84th percentile −1 to +8%) and the median Mg/Si by +14% (−1 to +32%; 6,256 spectra, Fig. 6). With the two-temperature XSM spectrum, the changes are −4% for Al/Si and −21% for Mg/Si (6,100 spectra). The per-month medians are listed in Table 5.

Two conclusions follow. First, Al/Si changes by at most a few per cent under either description of the measured solar spectrum, which supports its use as the more robust product. Second, Mg/Si depends noticeably on the assumed thermal structure of the corona: the isothermal and two-temperature descriptions of the same XSM spectra shift it in opposite directions. Both shifts lie within the ±20% temperature systematic stated in Section 9.5, but they show that this systematic domi-

Table 5: Median change (%) in Mg/Si and Al/Si when CLASS spectra are refitted with XSM-derived solar spectra instead of the GOES isothermal spectrum.

	XSM isothermal		XSM two-temperature	
Month	Mg/Si	Al/Si	Mg/Si	Al/Si
December 2024	+14	+3	−24	−6
June 2025	+15	+5	−15	−2
December 2025	+12	+2	−27	−6
All	+14	+4	−21	−4

nates the Mg/Si error budget. Neither description fits the XSM spectra perfectly (median reduced χ^2 of 3.9 and 2.8), plausibly because of variable coronal abundances, so this comparison constrains, rather than removes, the solar-spectrum systematic. A full XSM-based treatment of all months, ideally with free abundances, is a natural extension, and our model accepts any tabulated $S(E)$.

9.4. The instrumental aluminium component

The Al Kα excess in sunlit nightside spectra has a sharp onset near SOLARANG $\approx 93^\circ$ and disappears at the shadow boundary (Fig. 1). The most likely explanation is solar X-rays reaching aluminium in the instrument (filters, frames, housing; Pillai et al., 2021) when the Sun is behind the spacecraft. On the dayside the Sun is in front of the nadir-pointing instrument, and we expect the component to be negligible.

Its treatment affects Al/Si appreciably. Subtracting a month-averaged nightside spectrum, as in our earlier processing, removes a time-variable amount of Al. Fitting its amplitude freely, as in our alternative model, allowed it to absorb part of the lunar Al signal in some months; Al/Si reliability then dropped from 0.97 to 0.42 and the GRS correlation from 0.72 to 0.35. Other XRF instruments with aluminium structures or filters could show a similar effect, which the same nightside test would reveal.

9.5. Limitations

1. *Approximate solar spectrum.* The solar spectrum is isothermal at the GOES temperature. In synthetic tests, temperature errors of +30% and −20% bias Mg/Si by −15 to −13% and +28 to +33%, and Al/Si by +3 to +5% and +5 to +7%. Scaling the GOES temperature of real data by 0.8 and 1.2 changes Mg/Si by −25% and +27%, and Al/Si by −9% and +8% (Section 5). The XSM comparison for three months (Section 9.3) gives changes within these ranges. Ca/Si, Ti/Si and Fe/Si are not retrieved with this approach.
2. *Relative, not absolute.* The products are weight ratios to Si. No absolute calibration is attempted, and the amplitude A absorbs the solar flux, area and geometry.
3. *Detector efficiency.* The published filter transmission is applied. The SCD quantum efficiency and any surface dead layer are not modelled, because we found no published curve. This mainly affects the Si line relative to Mg and Al, and may explain why the median Al/Si (0.92) is higher than most returned soils. Such an offset would rescale the maps but leave their pattern and correlations unchanged.
4. *Physical effects not modelled.* Particle-size and roughness effects, phase-angle dependence beyond the plane-parallel geometry, and the angular dependence of scattering are not modelled.
5. *Spatial resolution.* Footprints are 25×37 km. The maps are 2° and should not be interpreted at finer scales.
6. *Small validation sample.* Nine landing sites, sampled at the scale of a footprint, give a test with little statistical power. The GRS comparison is limited by the GRS resolution and by its own model dependence for Mg and Al.

10. Conclusions

We have described a CLASS processing chain that uses only public data. It combines a simplified fundamental-parameter model, a CHIANTI solar spectrum at the GOES temperature of each exposure, the published filter transmission, a particle-background template and a Poisson likelihood fit with per-spectrum uncertainties and quality flags. Applied to 1,515,239 Level-1 exposures from 24 monthly archives, it gives 578,530 quality-controlled spectra and global 2° maps of Mg/Si and Al/Si with complete observed coverage. Our main findings are:

1. Element/Si ratios depend appreciably on the incident solar spectrum. Solar line emission carries most of the flux that excites Mg, Al and Si, and an isothermal spectrum at the GOES temperature brings the retrieved Mg/Si into the lunar range.
2. CLASS nightside spectra contain a solar-driven instrumental Al Kα component when the Sun illuminates the spacecraft from behind. How it is treated in the dayside background model affects Al/Si.
3. With a GOES-based solar spectrum, Mg/Si and Al/Si can be recovered, whereas Ca/Si, Ti/Si and Fe/Si are dominated by solar-spectrum systematics.
4. Al/Si agrees with Lunar Prospector GRS ($r = 0.72$) and with nine landing-site soils ($r = 0.81$), and several maria appear as distinct units. Mg/Si agrees with GRS ($r = 0.61$); its landing-site comparison is inconclusive, as it is for GRS itself.
5. For three months, replacing the GOES-based spectrum by XSM-derived spectra changes the median Al/Si by a few per cent and Mg/Si by −21 to +14%, depending on the assumed thermal structure. This supports Al/Si as the more robust product and identifies the solar spectrum as the main uncertainty for Mg/Si.

The code, mapping-set results and maps are public. Extending the XSM-based treatment to all months, with free coronal abundances, is the most direct way to reduce the remaining systematic uncertainty.

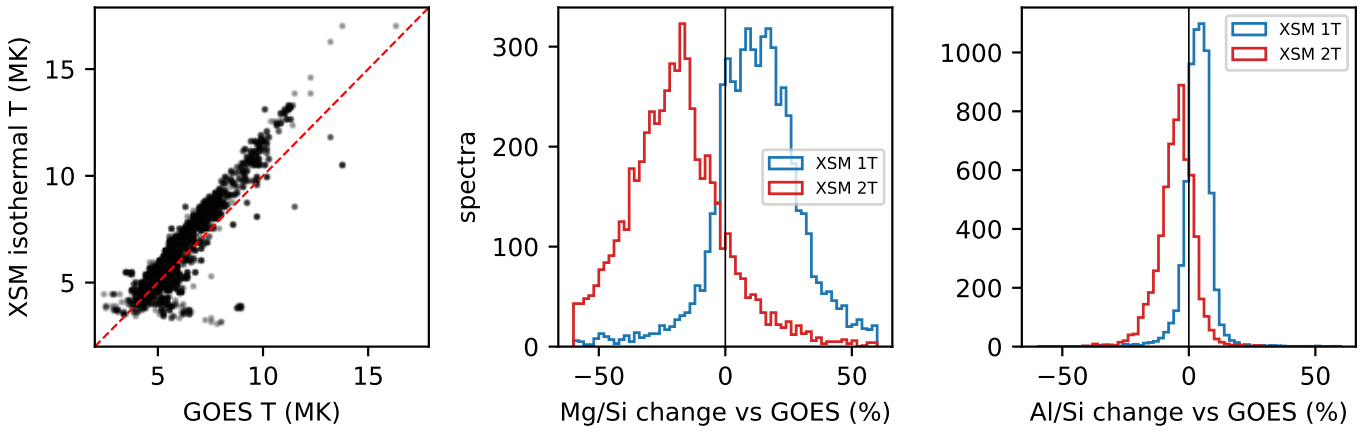


Figure 6: GOES- versus XSM-based solar spectra. Comparison for 6,327 CLASS exposures in December 2024, June 2025 and December 2025. Left: XSM isothermal temperature against GOES temperature (dashed: equality). Middle and right: change in Mg/Si and Al/Si when the CLASS spectra are refitted with the XSM isothermal (1T) or two-temperature (2T) spectrum instead of the GOES one.

Appendix A. Per-month statistics and calibration

Table A.6 lists the per-month exposure counts, mapping-set size, fitted energy-scale parameters and the source of the particle template.

Table A.6: Per-month exposure counts, mapping-set size, fitted energy-scale parameters and particle-template source.

Month	Files	Dayside	Mapped	g (eV/ch)	o (eV)	$\mathrm{FWHM}_{1.49}$ (eV)	Umbra template
2020-12	3,931	3,051	94	13.379	29.3	107	own
2021-01	4,634	3,501	0	13.383	13.7	108	own
2021-02	40,247	27,555	546	13.896	-24.4	92	own
2022-03	5,896	2,793	731	13.717	1.1	119	Dec 2024
2024-03	15,076	7,939	1,982	13.678	8.6	119	Dec 2024
2024-11	104,125	74,962	51,815	13.759	1.1	165	own
2024-12	97,345	75,720	42,704	13.737	2.2	185	own
2025-01	51,298	38,780	27,580	13.740	3.1	162	own
2025-02	92,570	63,780	44,826	13.739	3.6	156	own
2025-03	40,571	20,519	4,606	13.765	0.7	145	own
2025-04	81,585	48,469	18,672	13.839	-6.5	145	own
2025-05	94,521	65,910	39,426	13.634	16.1	156	own
2025-06	97,669	68,095	48,494	13.819	-5.8	168	own
2025-07	90,614	67,190	44,079	13.641	15.7	161	own
2025-08	97,593	66,706	39,325	13.821	-5.9	156	own
2025-09	40,747	23,210	11,039	13.898	-13.2	151	own
2025-10	27,237	16,137	2,018	13.684	9.4	169	own
2025-11	105,203	74,667	41,296	13.700	8.0	169	own
2025-12	92,318	71,126	44,952	13.747	3.3	176	own
2026-01	85,725	65,896	43,891	13.856	-8.1	176	own
2026-02	84,922	61,436	24,301	13.738	7.0	156	own
2026-03	31,184	18,295	2,913	13.669	15.1	160	own
2026-04	29,064	16,453	5,528	13.821	0.8	148	own
2026-05	101,164	71,799	37,712	13.766	7.9	155	own

Data availability

CLASS Level-1 data (ISRO Science Data Archive, 2026a) and XSM data (ISRO Science Data Archive, 2026b) are available from the ISRO PRADAN portal. GOES XRS 1-minute science-quality data are available from NOAA/NCEI (NOAA National Centers for Environmental Information, 2026). The Lunar Prospector GRS elemental abundances are available from the PDS Geosciences Node (Prettyman, 2012). The processing code, the per-footprint results of the mapping set, the maps and the scripts that generate every number and figure in this paper are available at `https://github.com/GeeteshPaidi/SomaDhatuNirupana` and archived on Zenodo (Paidi et al., 2026).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to assist with code development, data analysis, and drafting and editing of the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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